

My research focuses on artificial intelligence, machine learning, and computational modelling for complex high-dimensional datasets, with applications ranging from biomedical imaging to data-driven scientific and technological systems. My [doctoral dissertation](#) was nominated as a [distinguished PhD thesis](#) by Heidelberg University, Germany and was later published as the book "[Chromatin Architecture](#)" by Springer Nature. I currently have a [H-index of 17 with more than 20 peer-reviewed publications](#) and over [1400 citations](#) in journals such as [Nature Methods \(2x\)](#), [PNAS](#), [Nature Neuroscience](#), and [Nature Reviews](#), among others. My research contributions have been recognised internationally through awards, invited lectures, and collaborations across multiple institutions. I am an [\(elected\) Fellow of the Royal Microscopical Society](#) and actively contribute to the scientific community as a journal reviewer, scientific editor, and panel expert for funding agencies. To date, I have secured over \$1M in research funding. As a computational scientist and tool developer, I am particularly interested in developing [AI-driven approaches for analysing complex datasets](#), including large-scale imaging and biological data, with broader applications across data science, machine learning, and computational modelling. Alongside my research, I am deeply committed to teaching and mentoring students. I have taught a wide range of subjects including programming (R, Python, MATLAB), machine learning, statistics, mathematical modelling and image analysis, as well as delivering interdisciplinary courses and STEM outreach activities.

Personal Details

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Current [Medicine and Healthcare Regulatory Agency \(MHRA\)](#), Senior Scientist, UK, Permanent position I am establishing a causal-inference and AI/ML pipeline to quantify the risk of antibiotics

Education

- 2013-2016 PhD, Ruprecht-Karls-Universität Heidelberg, Heidelberg, Germany, [Springer Best PhD Theses Prize](#)
- 2009-2012 Master of Science (Tech.), Aalto University (with University of Helsinki), Finland, Distinction
- 2005-2009 Bachelor of Technology, Uttar Pradesh Technical University, India, First Division

Research & Industrial Experience

- 2024–2025 [TU Delft and University of Edinburgh](#), Senior Research Fellow, UK and Netherlands
We compiled 200 TB of digital pathology data and are developing foundation AI models for cancer imaging-based markers ([bioRxiv 2024](#)). With [Carlas Smith](#), we are also developing the Bayesian Van Trees inequality to improve the quantification of localization precision for superresolution microscopy ([JMI 2024](#)).
- 2022–2024 [Institute of Cancer Research \(ICR\) and Royal Marsden Hospital](#), AI/ML Team Lead, UK
Led development of several deep learning pipelines for digital pathology, including rhabdomyosarcoma tissue phenotyping with [Janet Shipley \(80 slides\)](#) and CD31 [vessel segmentation](#) with [Emma Harris](#) and [Navita Somaiah](#). With [Lothar Schermelleh](#), we highlighted several open challenges in superresolution microscopy ([JCS 2025](#), [Nat Rev 2024](#), [PTRSA 2022](#)) and co-organized the [Royal Society SIMposium](#) conference.
- 2020–2022 [National Physical Laboratory \(NPL\)](#), Higher Research Scientist, UK
Pioneered structure-based modeling for MINFLUX data ([Nat Methods 2023](#); [Phil Trans A 2022](#)) and promoted open science with [Johannes Hohlbein \(Nat Methods 2022\)](#). I helped organize [Royal Society Open Microscopy meeting](#) and developed a framework for correlative cryo-EM and superresolution microscopy with [Linkam](#).
- 2018–2020 [University of Cambridge & Cambridge Stem Cell Institute](#), Research Associate, UK
In [Rowitch lab](#), I set up a confocal-based spatial transcriptomics/RNAscope pipeline to quantify single-cell gene expression ([Nat Neurosci 2020](#)). I extended this work in the [Klenerman lab](#) for chromatin imaging and developed a machine learning pipeline to link morphology to molecular markers. I also set up a HPC cluster.

- 2017–2018 [University of Oxford & Oxford Nanoimaging \(ONI\)](#), Visiting Scientist, UK
Contributed to the development and sales of [Nanoimager](#) and integrating structured illumination microscopy.
- 2016–2017 [Carnegie Institution & Institut Curie](#), Research Associate, UK
Pioneered a laser-free super-resolution microscopy approach, enabling correlative imaging with STED microscopy on same fields-of-view and sample preparations ([Phil Trans A 2021](#); [JMI 2024](#)).
- 2013–2016 [IMB & Max Planck Institute for Polymer Research](#), Research Assistant, Germany
Co-developed UV induced photo-conversion of DNA dyes for single-molecule imaging ([Nucleus 2014](#); [ECR 2015](#)); mapped epigenetic landscapes in meiosis and stress ([PNAS 2015](#); [Genome Biol 2015](#)).
- 2009–2012 [Aalto University, Tampere University & University of Helsinki](#), Research Assistant, Finland
My bachelors thesis focused on neural networks for the prediction of protein secondary structure, while my master's explored histone code using Boolean models ([GCB 2016](#); [Biosystems 2018](#); [Springer 2024](#)). At the [University of Helsinki](#), I worked on the annotation of the [Birch genome](#).

Awards & Achievements

I was elected as a [Fellow of the Royal Microscopical Society \(FRMS\)](#) in 2022 in recognition of my contributions to imaging innovation, open science, and community leadership. As a [Royal Microscopical Society \(RMS\) Ambassador](#), I actively support early-career researchers and promote interdisciplinary engagement across imaging, biology, and computation. I received the [Springer Best PhD Thesis Prize](#) (2016) for my doctoral research at Heidelberg University on superresolution imaging of DNA and chromatin architecture. My research has been recognised at international conferences, earning several best poster awards, including from EMBO (2015), JCB (2016), and ICCB (2016). I was an [Honour Student](#) at Aalto University (2010) for academic excellence in computer science.

Key Discoveries & Innovations

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|-------------------------|---|
| Microscopy | Discovered that superresolution microscopy can be performed with a conventional epifluorescence microscope setup and a non-coherent light source like a lamp, PTRSA |
| Nuclear pores | Developed mathematical modelling to study organisation of nuclear pores, Nature Methods |
| Histone code | Proposed the role of histone code in higher-order chromatin folding, GCB |
| Meiosis | First to characterise the high-resolution epigenetic landscape of mouse meiotic chromosomes, PNAS |
| Chromosomes | Co-discovered that DNA can reversibly adapt to new configurations under stress, Genome Biology |
| Microscopy | Established sequential STED and SMLM imaging on same sample and field-of-view that have contrasting sample and acquisition requirements, JoM |
| Photophysics | Found UV-induced photo-conversion of commonly used DNA dyes like DAPI, Nucleus |
| Spatial Transcriptomics | Set-up the RNAscope and developed a deep learning pipeline to link morphology information to molecular features, Nature Neuroscience |

Key Skills

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|-------------|--|
| Programming | Python (NumPy, SciPy, Pandas, Scikit-learn, PyTorch, TensorFlow), MATLAB, R |
| Microscopes | Home-built and commercial microscopes - Nikon, Aurox, Leica, Olympus and Perkin Elmer Operetta |
| Genomics | Next-generation sequencing analysis (ChIP-seq, MNase-Seq, HiC datasets) |
| Modeling | Boolean models, mathematical modeling, structure-based modelling |

Fellowship Applications

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| 2025 | Classification of chromosome topology using mathematical modelling and machine learning, BBSRC Fellowship, Wellcome CDA, University of Cambridge (Physics Department), £800K, Unsuccessful |
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- 2021 Building the cell nucleus design principles of chromosome folding, Royal Society University Research Fellowship, University of Cambridge (DAMTP), £1.5M, Unsuccessful

Grant Applications (Successful)

- 2025 [DeepSIM: Live Tissue Imaging at the Nanoscale](#), assisted Carlas Smith in grant preparation
- 2022 The Royal Society Grant for [Open Microscopy](#), assisted in grant preparation
- 2021 Successful funding and purchase for a confocal microscope at NPL, Lead applicant, £300K
- 2020-21 NPL M4R grants with Aurox Ltd, Linkam Stages and CoolLED, Lead applicant, £100K
- 2020 SIM Hackathon funding from Zeiss, Co-applicant, £10K, University of Cambridge, UK
- 2019 The Royal Society Grant for Hooke proposal [SIMposium](#), Lead applicant
- 2019 Next Generation Imaging workshop, Industrial funding, Lead applicant, £10K, CRUK, UK
- 2018 Funding and setup of a confocal microscope, Lead applicant, £350K, University of Cambridge, UK
- 2018 Initiated ITN Horizon 2020 grant, Main coordinator for ONI, Oxford, UK
- 2017 [The organization of animal cell nuclei](#), assisted Joe Gall in grant preparation
- 2016 Several travel grants including EMBO and ASCB

Teaching & Training Experience (selected)

- 2025 Statistical Signal Processing (TU Delft), substitute course lecturer on behalf of Carlas Smith
- 2024 [ESRIC summer school](#), Invited talk and lecturer for Hands-on Practical Workshop
- 2024 [Hands-on superresolution workshop](#), University of Leicester, Lecturer (invited)
- 2023 [Transforming Health Services with AI](#), Coventry University, London, Lecturer (invited)
- 2023 Machine learning and data analysis, ICR, London, Course organiser and lecturer
- 2020 [The STEM hub outreach talks](#), NPL, London. [Microscopy Wonderland](#), Lecturer
Delivered 10 talks to 60 schools in 5 days, approximately 4500 pupils participated
- 2017 [Modelling of Biological Systems](#) at JGU, Mainz, Lecturer, Summer term
- 2015 [Modelling with Matlab in Optical Imaging](#) at IMB, Mainz, Teaching Assistant
- 2015 [Advanced Seminar: Biophysics of Genome Structure](#) at JGU, Mainz, Lecturer
- 2014-15 [Basics of Biomedical Optics 1 and 2](#) at IMB, Mainz, Teaching Assistant
- 2014-15 [Medical Physics Program](#) at University of Heidelberg, Teaching Assistant
- 2014-2015 [Fundamental Physical Problems in Optical Imaging](#), Lecturer and Teaching Assistant

Mentorship

- 2024-25 TU Delft, the Netherlands, supervised 3 MSc students, 3 PhD students on behalf of Carlas Smith
- 2022-24 Institute of Cancer Research, Sutton, UK, led the ML/AI team, supervised 2 analytical scientists
- 2018-19 University of Cambridge, Cambridge, UK, supervised 1 summer student, 3 BSc/MSc students
- 2013-16 IMB/JGU, Mainz, Germany, supervised 3 summer students, 5 BSc/MSc students

Conference Organization

- 2025 [ORBIT: Organisation of Genome in Space and Time](#), TU Delft, Lead organiser
- 2022 [SIMposium conference](#), [Royal Society Theo Murphy Meetings](#), London, UK, Lead organiser
- 2020-2023 [Imaging ONEWORLD series](#), Cambridge, UK, Lead organiser
- 2019 [Next Generation Imaging \(NGI\) workshop](#), CRUK, Cambridge, UK, Lead organiser

Scientific Editor & Reviewing Activities

- 2025 [Journal of Microscopy](#), Microscopy at a Glance poster article series, Lead Scientific Editor (ongoing)
- 2025 [Fonds Wetenschappelijk Onderzoek, Vlaanderen \(FWO\) expert panel](#), Molecular Cellular Biology
- 2023-24 [Journal of Microscopy](#), Imaging ONEWORLD poster article series, Lead Scientific Editor
- 2023 [Applied Physics Letters](#), Advances in Optical Microscopy for Bioimaging, Lead Scientific Editor
- 2022 [Philosophical Transactions of the Royal Society](#), Lead Scientific Editor, SIMposium, Part 2
The most cited and downloaded Royal Society issue in the year 2022, 2023
- 2021 [Philosophical Transactions of the Royal Society](#), Lead Scientific Editor, SIMposium, Part 1
- 2020- Regular reviewer for journals including Nature Communications, PNAS, APL, Scientific Reports, Communications Biology, Journal of Microscopy, and Philosophical Transactions

Committee Memberships (selected)

- 2023 CEO advisory board, Institute of Cancer Research, UK, Staff scientist representative
- 2022- European Microscopy Society and Royal Microscopy Society, Member
- 2018-2019 The International Society of Code Biology (ISCB), Governing Board
- 2016-2017 American Society for Cell Biology, European Society for Molecular Imaging, Member
- 2015- Microscopy, Imaging, Bioinformatics and Machine learning consultancy

Talks at Summer Schools, Workshops and Conferences

- 2025 [Foundations of Causal Inference](#), Isaac Newton Institute for Mathematical Sciences, Cambridge
- 2025 [Physics of Life Summer School](#), Invited talk
- 2025 [Southampton Imaging Conference](#), Illuminating All Things Big and Small, Invited talk
- 2024 [Physics of Biological Cells](#), Higgs Centre for Theoretical Physics, Edinburgh, UK
- 2024 [Microscopy Technology Dissemination to Underserved Communities](#), Janelia, US
- 2021 [Setting up rules for "Good Nanoscopy Practice](#), ICON EUROPE, Podium Discussant
- 2021 [Assessment of 3D MINFLUX data for quantitative structural biology](#), ICON EUROPE
- 2019 Dynamic in-situ microscopy relating structure and function, The Royal Society, London
- 2019 [The beautiful higher-order structures of DNA](#), 6th Conference in Code Biology, Keynote talk
- 2019 [DAMTP BioLunch series](#), University of Cambridge, Cambridge, UK
- 2017 Istituto Italiano di Tecnologia, Department of Nanophysics, [Leonardo lecture](#), Genoa, Italy
- 2016 Stanford University, Department of Development Biology, Palo Alto, US

Public Engagement & Outreach in Biophysics

My public engagement work in biophysics is driven by a simple idea: the invisible world of cells, DNA and light should be accessible to anyone who is curious, whether they are school pupils, researchers in low-resource settings, or members of the public encountering microscopy for the first time.

Schools & young audiences: Through my [Microscopy Wonderland](#) STEM Hub outreach talks, I designed and delivered an intensive programme introducing fluorescence microscopy, DNA and cells to school pupils using accessible language and live demonstrations; over 5 days gave 10 talks to 60 schools, reaching ~4,500 pupils and encouraging questions about “invisible worlds”.

Open science & accessible microscopy: I have worked to lower technical and financial barriers to advanced imaging by sharing hardware designs, acquisition workflows and analysis code. This includes co-authoring a [Nat. Methods](#) perspective on open hardware, developing a laser-free super-resolution method that enables localisation microscopy on conventional epifluorescence microscopes ([Phil. Trans. R. Soc. A](#)), and publishing practical, concept-

driven guides in [Nat. Rev. Mol. Cell Biol.](#) and [J. Cell Sci.](#) that address common misconceptions about super-resolution and resolution limits. Alongside these, I provide free training and informal consulting, often to researchers primarily in Eastern Europe, Asia and Africa, to help them adapt super-resolution to their existing microscopes using affordable, open-source tools. For me, open science is not just a principle; it is the practical route by which microscopy can benefit everyone, not only the best-funded laboratories.

Community-focused meetings & global online initiatives: I have created inclusive forums that blend cutting-edge biophysics with broad, accessible discussion. With Royal Microscopical Society (RMS), I started [Imaging ONEWORLD series](#), an online seminar programme that significantly broadened access to advanced microscopy for students and researchers worldwide. In 2020, I acquired funds and led the Royal Society [SYMposium](#) conference focused on super-resolution microscopy explicitly designed to foster open discussion of limitations and best practice. In 2019, I organised an in-person [Next Generation Imaging \(NGI\) workshop](#) at CRUK Cambridge, which brought together academic and industry speakers. Most recently, I organised [ORBIT: Organisation of Genome in Space and Time](#) at TU Delft (2025), an interdisciplinary meeting bringing together imaging, genomics and modelling communities.

Diversity, inclusion & underserved communities: As a researcher of South Asian heritage with an international career in Finland, Germany, the UK, the US and the Netherlands, I have seen first-hand how access to scientific opportunities is shaped by structural inequities and local culture. These experiences have motivated me to broaden participation by recognising diversity in its demographic, disciplinary and experiential dimensions. My interdisciplinary background in computer science and biology exemplifies the value of crossing boundaries, and I encourage students to treat such interdisciplinarity as a strength. I have supervised early-career researchers from underrepresented groups and underserved regions, with a particular focus on making advanced technologies such as AI and microscopy accessible. At Janelia's [Microscopy Technology Dissemination to Underserved Communities](#), I presented strategies for adapting super-resolution imaging workflows to existing microscopes in low-resource settings.

Research Summary

AI and machine learning: My academic journey started with developing machine learning algorithms to [predict secondary structure of proteins](#). Since then I have developed several algorithms that include [Boolean models](#) for my master thesis, data reconstruction, analysis and visualization pipeline for [single-molecule microscopy](#), image stitching, annotation, and machine learning pipeline for [high-throughput spinning disk confocal microscopy](#) datasets. Recently, my focus has shifted towards developing [deep-learning algorithms](#) for digital pathology and cell biology.

Chromosome biology: My fascination for science began with [histone code hypothesis](#), which states that small chemical modifications can lay the [foundational rules for DNA folding](#). For my masters thesis, I developed Boolean models to explore which histone modifications frequently co-occur, leading me to propose a role for [histone code in higher-order chromatin folding](#). As the pipeline was based on genomic datasets, it occurred to me to visualize the histone modifications along with DNA. Subsequently, for my Ph.D., I developed a new superresolution microscopy method to describe the first high resolution epigenetic landscape in single cells and [along individual chromosomes](#). As the next step, during my postdoctoral work at Cambridge, I set up high-content screening systems for the Human Cell Atlas project which I also used to study the histone code in [thick, large tissue sections](#).

Superresolution microscopy: During my Ph.D., we found that common DNA dyes such as DAPI can be photoconverted for [single-molecule imaging of DNA](#), which helped us to discover several [new chromatin domains](#). Following my Ph.D., I moved to Carnegie Institution, where I did not have access to superresolution microscopy. This challenge led me to develop a laser-free superresolution microscopy, enabling localization microscopy on any [conventional microscope](#). I also established a protocol that allowed [correlative STED and SMLM](#) studies on the same sample under identical imaging conditions. Around this time, a new wave of single-molecule techniques claimed Angstrom-level resolution with light microscopy. I critically examined these claims and developed [structure-based modeling](#) approaches to evaluate the precision and applicability of these methods. Over the years, I also contributed with several high impact [review articles](#), [perspectives](#) and have also been asked for my opinions by popular science magazines like [New Scientist](#).

Open Science and Microscopy Accessibility

I have been pushing open microscopy by sharing hardware designs, acquisition workflows, and analysis code to break long-standing barriers in the field. This includes co-authoring a [Nat. Methods](#) perspective on open hardware, developing a laser-free super-resolution method to cut costs ([Phil. Trans. R. Soc. A](#)), and publishing guides in [Nat. Rev. Mol. Cell Biol.](#) and [J. Cell Sci.](#) to tackle common superresolution misconceptions. I have also offered free training, often to researchers from Eastern Europe, India, and Africa, to help them adapt super-resolution to their existing microscopes. For me, open science is not just a principle: it is the only way to ensure that microscopy benefits everyone, not just the best-funded laboratories.

Diversity Statement

As a researcher of South Asian heritage with an international career in Finland, Germany, the UK, the US, and the Netherlands, I have seen how access to opportunities in science is shaped by both structural inequities and the cultural context. These experiences have motivated me to broaden participation by recognizing diversity in its many forms: demographic, disciplinary, and experiential. My interdisciplinary background in computer science and biology exemplifies the value of crossing boundaries, and I encourage students to view such interdisciplinarity as a strength. In my teaching and mentoring, I adapt to varied learning styles and backgrounds, ensuring inclusive assessments and career preparation. I have supervised students from underrepresented groups and underserved regions, with a particular focus on making advanced technologies such as AI and microscopy accessible. Beyond the classroom, I have supported outreach initiatives, organized inclusive scientific meetings, and contributed to professional societies that promote equity. Through these efforts, I aim to create research environments where diversity is not only represented but actively drives innovation and collaboration.

Teaching Statement

I am a dedicated and interdisciplinary educator with experience teaching across computer science, biomedical engineering, and health innovation. My teaching philosophy is grounded in active learning, problem-solving, and real-world application. I aim to cultivate critical thinkers who are not only technically proficient but also capable of applying their skills responsibly in complex biomedical and healthcare settings. I have developed and delivered courses using both in-person and digital platforms (e.g., Moodle, Aula) and have led curriculum development at TU Delft, Coventry University, and the Institute of Cancer Research (ICR). I also organised an online lecture series for the Royal Microscopical Society, promoting accessible microscopy education worldwide.

Teaching Approach and Philosophy: My approach is inspired by constructivist principles and strengthened through reflective practice, peer observation, and engagement in programs such as the Teaching Associates Program. I focus on stimulating curiosity, encouraging students to ask “why,” “how,” and “what if,” while guiding them through problem-solving frameworks. I integrate diverse methods, including:

- Concept summaries, applied case studies, and project-based learning.
- Peer-to-peer teaching and exam question creation to deepen engagement.
- Regular feedback loops, including tutorial customisation based on student input.

Teaching Goals

- Cultivate independent, interdisciplinary problem-solvers.
- Promote evidence-informed and ethically aware innovation.
- Connect computational thinking to human health outcomes.

Semester-Long Courses Delivered:

Stochastic Signal Processing for Biomedical Students (TU Delft): Focused on statistical modeling and signal estimation in noisy biomedical contexts, this course taught students how to apply stochastic methods such as Wiener filtering,

ARMA models, and spectral analysis to physiological signals and molecular motion. Emphasis was placed on interpreting uncertainty in real biomedical systems.

[Introduction to Machine Learning \(Institute of Cancer Research\)](#): This course introduced supervised and unsupervised learning, neural networks, and model evaluation with practical implementation in Python. Students applied these tools to biomedical imaging and health diagnostics, completing a final project using real-world datasets.

[Transforming Health Services Using AI \(Coventry University\)](#): This interdisciplinary course explored AI's role in reimagining healthcare delivery. Covering digital health transformation, personalized medicine, and service design, students developed AI-driven solutions for service improvement while critically examining ethical, economic, and policy implications. I designed all 15 lectures for this course.

[Cell Biology for Biomedical Students \(University of Mainz\)](#): This course provided a comprehensive foundation in modern cell biology, covering the structure, function, and dynamics of cellular organelles, cytoskeletal systems, and signal transduction pathways. Emphasis was placed on linking molecular mechanisms to cellular processes such as division, differentiation, and apoptosis. Students engaged in literature-based projects, critically analysing landmark discoveries and applying concepts to biomedical case studies.

[Microscopy and Image Analysis \(University of Cambridge\)](#): This hands-on course introduced the principles and applications of advanced light microscopy, including fluorescence, confocal, and super-resolution techniques. Students were trained in experimental design, sample preparation, and data acquisition, followed by computational analysis of imaging datasets using Python and MATLAB. A strong focus was placed on quantitative image analysis workflows, enabling students to extract biologically meaningful parameters from complex cellular images.

Top 5 Publications (since 2020)

#	Title	Authors	Journal (Year)	PMID	Citations	Impact Factor (2024)
1	Resolution in super-resolution microscopy definition, trade-offs, and perspectives	Prakash, Kirti*, David Baddeley, Christian Eggeling, Reto Fiolka, Rainer Heintzmann, Suliana Manley, Aleksandra Radenovic, Carlas Smith, Hari Shroff, Lothar Schermelleh	Nature Reviews MCB (2024)	38951703	15	128.7
2	Assessment of 3D MINFLUX data for quantitative structural biology in cells	Kirti Prakash*, Alistair Curd*	Nature Methods (2023)	36522506	27	51.7
3	Open microscopy in the life sciences: quo vadis?	Johannes Hohlbein, Benedict Diederich, Emmanuel G. Reynaud, Séamus Holden, Kirti Prakash	Nature Methods (2022)	36008630	56	51.7
4	Laser-free super-resolution microscopy	Kirti Prakash	Phil. Trans. R. Soc. A (2021)	33896204	15	4.7
5	Astrocyte layers in the mammalian cerebral cortex revealed by a single-cell in situ transcriptomic map	Omer Ali Bayraktar, . . . , Kirti Prakash, . . . , David Rowitch	Nature Neuroscience (2020)	32203496	453	20.0

References

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Publications list (*co-first/corresponding/last author)

- 2026 Korovin, ..., [Prakash](#) and Smith. Quantifying the spatio-temporal resolution. [bioRxiv](#)
- 2026 Spengler, Korovin, [Prakash](#) et al. CLEAR-IT. [Communications Biology](#)
- 2025 [Prakash*](#) et al. Resolution in superresolution microscopy (Part 2). [Journal of Cell Science](#)
- 2024 [Prakash*](#) et al. Resolution in superresolution microscopy (Part 1). [Nature Reviews MCB](#)
- 2024 [Prakash*](#) & Fournier. DNA and Chromatin Structure in Epigenetic Signaling. [Springer](#)
- 2024 Smith, Kalisvaart & [Prakash*](#). Unveiling the limits of precision in MINFLUX. [Journal of Microscopy](#)
- 2023 [Prakash*](#) & Curd*. Assessment of 3D MINFLUX data. [Nature Methods](#)
- 2023 [Prakash*](#) et al., Opinions on imaging and cutting-edge developments in light microscopy. [APL](#)
- 2023 [Prakash](#). Correlative superresolution microscopy with deep UV reactivation. [Journal of Microscopy](#)
- 2023 Reichelt, ..., & [Prakash*](#) Imaging ONEWORLD. [infocus Magazine](#)
- 2022 [Prakash*](#) et al. Super-resolution microscopy: a brief history and new avenues. [Phil Trans Royal Soc A](#)
- 2022 Hohlbein, ... & [Prakash*](#). Open microscopy in the life sciences: quo vadis?. [Nature Methods](#)
- 2022 [Prakash](#). At the molecular resolution with MINFLUX?. [Phil Trans Royal Society A](#)
- 2021 [Prakash](#). Laser-free super-resolution microscopy. [Phil Trans Royal Soc A](#)
- 2021 [Prakash*](#) et al. Super-resolution structured illumination microscopy. [Phil Trans Royal Soc A](#)
- 2020 Bayraktar, ..., [Prakash](#), et al. Transcriptomic map of astrocyte layers. [Nature Neuroscience](#)
- 2018 [Prakash*](#) & Fournier*. Implication of the histone code in building the genome structure. [Biosystems](#)
- 2017 [Prakash*](#) & Fournier. Histone code and higher-order chromatin folding: a hypothesis. [GCB](#)
- 2017 [Prakash*](#). Chromatin Architecture. [Springer Nature](#)
- 2016 Zurek-Biesiada, Szczurek, [Prakash](#) et al. Quantitative localization microscopy. [Data in Brief](#)
- 2016 Zurek-Biesiada, Szczurek, [Prakash](#) et al. Localization microscopy of DNA in situ. [ECR](#)
- 2015 Kirmes., Szczurek., [Prakash](#), et al. Reversible compaction of chromatin. [Genome Biology](#)
- 2015 [Prakash](#) et al. Periodic patterns of chromatin along pachytene chromosomes. [PNAS](#)
- 2014 Szczurek*, [Prakash*](#), Lee*, et al. Single-molecule localization microscopy of DNA. [Nucleus](#)

Opinions/comments in popular science magazines/journals

- 2026 [Nature](#), Making samples one billion times bigger lets simple microscopes pinpoint amino acids
- 2024 [New Scientist](#), Tiniest 'ruler' ever measures distances as small as an atom's width

Publication summary

Summary >20 Journal papers, 1 Book, 5 Book chapters, >10 conference papers

Author stats Citations >1500, h-index - 17, i10-index -19

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